Python Random Module: Dice, Games and Surprises

Make your programs unpredictable and fun.

Python | Beginner | Ages 9–16

💡 The Big Idea

- The random module lets Python make choices by chance, like rolling a dice or shuffling cards.
- You must import it first: `import random`.
- Almost every game uses randomness: dice, loot, enemies, quizzes.
- Computers use pseudo random numbers, which look random but follow a pattern.

🎲 The Functions You Need

- `random.randint(1, 6)` gives a whole number from 1 to 6. Both ends are included.
- `random.random()` gives a decimal from 0 up to (not including) 1.
- `random.uniform(1, 10)` gives a decimal between two numbers.
- `random.choice(items)` picks one item from a list.
- `random.shuffle(items)` mixes up a list in place.
- `random.sample(items, 3)` picks 3 different items.
- `random.randrange(0, 10, 2)` picks from a range with a step.

🌱 Seeds: Repeatable Randomness

- `random.seed(42)` starts the generator at a fixed point.
- The same seed gives the same numbers every time. Great for testing and fair competitions.
- Without a seed, Python starts differently every run.
- The examples on this sheet use seeds so the output matches yours.

🎮 Ideas for Projects

- Dice roller and coin toss.
- Number guessing game.
- Rock, paper, scissors against the computer.
- Random quote or joke generator.
- Password generator and a lucky draw picker.

💻 Code Examples

Roll a dice python

```python
import random
random.seed(7)
for roll in range(5):
    print("You rolled", random.randint(1, 6))
```

▶ **Output:**

```
You rolled 3
You rolled 2
You rolled 4
You rolled 6
You rolled 1
```

seed(7) makes the five rolls repeat exactly the same way each time.

Pick from a list python

```python
import random
random.seed(3)
colours = ["red", "green", "blue", "yellow"]
print(random.choice(colours))
print(random.sample(colours, 2))
```

▶ **Output:**

```
green
['green', 'yellow']
```

sample() never picks the same item twice.

Shuffle a list python

```python
import random
random.seed(1)
cards = ["A", "K", "Q", "J"]
random.shuffle(cards)
print(cards)
```

▶ **Output:**

```
['J', 'A', 'Q', 'K']
```

shuffle changes the original list.

Decimals python

```python
import random
random.seed(5)
print(round(random.random(), 3))
print(round(random.uniform(10, 20), 2))
```

▶ **Output:**

```
0.623
17.42
```

round() keeps the output tidy.

Coin toss python

```python
import random
random.seed(11)
result = random.choice(["Heads", "Tails"])
print(result)
```

▶ **Output:**

```
Tails
```

A tiny program with a big effect.

Number guessing game python

```python
import random
random.seed(4)
secret = random.randint(1, 10)
guess = int(input("Guess 1 to 10: "))
if guess == secret:
    print("You got it!")
elif guess < secret:
    print("Too low. It was", secret)
else:
    print("Too high. It was", secret)
```

▶ **Output:**

```
Guess 1 to 10: 5
Too high. It was 4
```

Combines random with input and if / elif / else.

📚 Key Terms

Module
A file of ready made Python tools.

import
Brings a module into your program.

random
The module for chance and choice.

randint
A random whole number between two values.

choice
Picks one item from a list.

shuffle
Mixes the order of a list.

seed
A starting number that makes random results repeatable.

Pseudo random
Looks random but follows a formula.

Inclusive
Includes both end numbers.

Sample
A group of different items picked from a list.

🧮 Quick Reference

```
import random
random.randint(1, 6)
random.random()
random.uniform(1.5, 4.5)
random.choice(items)
random.shuffle(items)
random.sample(items, 3)
random.seed(42)
```

✏️ Worked Examples

Q: Simulate rolling two dice and adding them.

A: d1 = random.randint(1, 6), d2 = random.randint(1, 6), print(d1 + d2). The total is between 2 and 12.

Q: Pick a random day of the week.

A: days = ["Mon", "Tue", ...]. print(random.choice(days)) picks one.

Q: Make a random 4 digit PIN.

A: pin = "". Loop 4 times adding str(random.randint(0, 9)) to pin. Print pin.

🚫 Common Mistakes

- Forgetting `import random` at the top.
- Naming your own file random.py, which breaks the import.
- Expecting `randint(1, 6)` to stop at 5. It includes 6.
- Using `random.shuffle` on a string. Use a list.
- Forgetting `shuffle` returns nothing, so `x = random.shuffle(items)` gives None.

👨‍👩‍👧 Parent Corner

- Roll a real dice 20 times, then make Python do it. Compare how often each number appears.
- Ask: If I roll a dice, what are the chances of a 6?
- Help them build a guessing game and ask for hints that tell too high or too low.
- Explain that games feel fair because of randomness.
- Challenge: add a limit of 5 guesses.

🎯 Quick Check

1. What must you write first to use random?

✅ import random.

2. Which numbers can random.randint(1, 6) give?

✅ 1 to 6, including both ends.

3. How do you pick one item from a list?

✅ random.choice(items).

4. What does random.seed(42) do?

✅ It makes the random numbers repeat the same way each run.

5. Does random.shuffle return a new list?

✅ No. It mixes the original list and returns None.